

Coding & Solution

Date	4 July 2026
Team ID	
Project Name	Comprehensive measure of well-being
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/cherish-codes/Global-WellBeing-HDI-Predictor
Programming Language(s)	Python, HTML5, CSS3, JavaScript
Framework(s) Used	Flask (Python backend micro-framework), Bootstrap (Frontend layout)

Key Features Implemented	Data Sanitization Engine: Dynamic missing value and empty space imputation for HDI.csv source files. Predictive Core: Serialized Machine Learning model execution via HDI.pkl to compute a bounded well-being output. Interactive Analytics Portal: Minimalist, uncluttered web form layout for real-time policy scenario input simulation.
Pending / Incomplete Features	Automated PDF Exporter: Directly generating and downloading a summary planning report of the predicted index results. Multi-Model Toggle: Switching between multiple algorithm backends on the front end.
Setup / Run Instructions	1. Clone the version repository and navigate into the root directory. 2. Install all required statistical dependencies via terminal command: <code>pip install flask scikit-learn pandas numpy</code> 3. Run the application instance locally: <code>python app.py</code> 5. Open your web browser and load the layout portal view at: http://127.0.0.1:5000

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

